Bhumi Lipi · Changelog & Update Report

Complete Summary of System Enhancements, UI Refinements & Hackathon Alignment

LandTech Innovation Hackathon 2026 · Round 1 Technical Submission Report · Track: Digital Land Records & Verification

Project & Team	**Bhumi Lipi** | Team Leader: **Mohammad Abrar Hasnat** (DIU ID: 253-15-689)
Development Scope	Frontend Polish, Architectural Compliance, Concept Note 2-Page Strict Alignment, Security & Verification Tests
Verification Baseline	**53/53 Automated Unit Tests Passing** | End-to-End Playwright Browser Walkthrough Clean | 0 Regressions

1. Executive Summary of Changes

Following the release of the official Round 1 Concept Note specifications, the Bhumi Lipi platform underwent comprehensive technical refinements across UI symmetry, typography, accessibility compliance, documentation synchronization, and code cleanliness.

- **Official 2-Page Concept Note Synchronization:** Updated `docs/CONCEPT_NOTE.md` and regenerated `docs/Bhumi_Lipi_Concept_Note.pdf` into a strict 2-page A4 document matching all 8 hackathon sections verbatim.
- **UI Symmetry & Card Alignment:** Solved visual imbalance on the Integrity view by introducing matching monospace hash inputs (84px height), synchronized subtitle heights, and horizontally aligned action rows.
- **Typography & Contrast Overhaul:** Enlarged table headers to 16px (weight 800) and data cells to 15px. Replaced low-contrast gold badges with high-contrast accessible `.flow-pill` components across light, dark, and high-contrast modes.
- **AI Disclosure Placement:** Clarified AI assistance strictly in documentation (`README.md`, `docs/CONCEPT_NOTE.md`) as mandated by contest rules, while removing redundant AI credits from the citizen-facing interface.

2. Frontend Architecture & Visual Innovations

Component / View	Enhancements Implemented
Integrity & Public Anchor (/#/integrity)	• Converted single-line input into an 84px dual-line `textarea.anchor-hash-box` to show full 64-char SHA-256 hashes without clipping. • Enforced matching 48px min-height subtitles and flex `justify-content: space-between` on both cards. • Horizontally aligned `[Copy]` and `[Verify proof]` buttons along the bottom baseline. • Positioned the interactive proof verdict banner as a full-width result below the 2-column grid.
About & Lifecycle (/#/about)	• Implemented the **4-Stage Countrywide Implementation Plan** (Pilot, District, Division, National) as an interactive bilingual table. • Upgraded the 5-step lifecycle pipeline with accessible `.flow-pill` design. • Retained comprehensive international comparison across 5 nations (Georgia, India, Rwanda, Estonia, Zambia).
Verify & Chain Inspection (/#/verify)	• Character-level diff highlight between expected and actual deed hashes. • Avalanche effect visualizer demonstrating how a 1-character tamper flips ~50% of hash bits.
Typography & Tables	• Upgraded `.t th` table headers to 16px (bold weight 800) and `.t td` to 15px in `web/css/app.css`. • Loaded dedicated Roboto Mono and Space Mono typefaces for all hashes, seal codes, and khatian numbers.
Accessibility & Responsive Design	• Supported text zoom (100%, 125%, 150%) with dynamic `calc()` token scaling. • Mobile bottom navigation bar for viewports < 768px. • High-contrast mode with crisp 2px black borders and zero text underlines.

3. File-by-File Technical Modification Matrix

File Path	Change Type	Detailed Description of Technical Changes
`docs/Bhumi_Lipi_Concept_Note.pdf`	**Regenerated**	Strict 2-page print-ready submission PDF matching the official concept note structure, tables, and pagination.
`docs/CONCEPT_NOTE.md`	**Updated**	Complete markdown copy of the 8-section concept note with problem statement P2, fraud P3, mutation P4, citizen check P5.
`README.md`	**Updated**	Updated problem statements, added 4-stage rollout plan table, official document links, and team leader details.
`web/js/views/integrity.js`	**Refactored**	Symmetric cards alignment, 84px monospace textarea input, Enter-to-submit key handler, full-width proof banner.
`web/css/app.css`	**Updated**	Table font enlargement (16px/15px), `.anchor-hash-box` responsive rules (84px desktop, 104px mobile), `.flow-pill` styles.
`web/js/views/about.js`	**Updated**	Integrated the 4-stage implementation table (পাইলট, জেলা, বিভাগ, জাতীয়) and high-contrast flow pills.
`web/js/app.js`	**Refactored**	Added `?v=1.0.1` query parameter to route imports for cache-busting; updated error boundary reload logic.
`web/js/views/check.js`	**Updated**	Eliminated text-border collision on chain cards by restoring generous internal padding and robust typography.
`web/js/components/seal.js`	**Feature**	SVG Seal with dynamic states: Genuine (green), Outdated (amber), Forged (red), and rotating cryptographic borders.
`web/js/components/chain-view.js`	**Feature**	Visual chain link renderer with broken-link crack visualization and tamper detection alerts.

4. Compliance & Verification Milestones

Automated Unit Tests	**53 Passed out of 53** across cryptography, lifecycle, mutation state machine, rules engine, and verifier scripts (Execution time: ~3.2 seconds).
Browser Automation (E2E)	Playwright end-to-end tests verify QR code rendering, theme toggles, modal dialogs, and verification flows.
Security Architecture	Strict Content-Security-Policy (CSP), textContent-only DOM rendering (zero XSS), append-only SQLite triggers, rate-limited JSON API.
Independent Verification	Includes standalone Python script (`tools/verify_anchor.py`) enabling third-party audit without server dependencies.

5. Summary of Deliverables Ready for Submission

- 📄 **Concept Note PDF:** `docs/Bhumi_Lipi_Concept_Note.pdf` (Strict 2-page official submission PDF).
- 📄 **Technical Changelog PDF:** `docs/Bhumi_Lipi_Changelog_and_Updates.pdf` (This 2-page executive summary).
- 📄 **Master Architectural Plan:** `docs/Bhumi_Lipi_Master_Plan.pdf` (Detailed 10-page technical specification).
- 💻 **Source Code Repositories:** Clean master branch with 0 uncommitted files, 53 tests passing, and Docker/Render configs.
- 📦 **Submission Archives:** `all_updated_files.zip` and `submission_and_updated_files.zip`.

Team Declaration: All land data in Bhumi Lipi is 100% synthetic. No real citizen names, NIDs, or live government records were used. AI coding assistants (Claude, Antigravity) were used for development and are fully disclosed in README.md as per rulebook guidelines.